

LEARN PYTHON PROGRAMMING – BEGINNER TO MASTER

Section: 7. String and its Methods

Lecture: 72. String Class

Section 1: Lecture Summary

This lecture introduces the concept of strings in Python from the perspective of classes and objects. It starts with a review of strings as data types, then explains that every string in Python is an object of a built-in class called `str`. To clarify the notions of class and object, the instructor draws an analogy with civil engineering: a class is like a blueprint or plan for a house, while objects are the actual houses built following that plan. The class groups related data and operations (methods) on that data. For the string class, data includes the actual string and its length, while operations include various string manipulation methods. The lecture demonstrates how to inspect the type of a string object and explore all available string methods using IDE features and the built-in `dir()` function. It ends with a promise to cover specific methods of the string class in upcoming lectures.

Section 2: Key Concepts and Explanations

- What is a String in Python?

- A string is a sequence of characters enclosed in single or double quotes.
- Example: `"Hello"`, `"Welcome"`.

- Everything in Python is an Object

- All data types in Python, including strings, are objects.
- Each object belongs to a class which defines its structure and behavior.

- Class vs Object (Civil Engineering Analogy)

- Class: The blueprint or plan for an entity (e.g., a house plan).

- Object: An instance created based on that blueprint (e.g., an actual house).
- Multiple objects can be created from the same class blueprint.

- Class Structure

- A class contains:
 - Data (attributes/variables) relevant to the entity (e.g., the string content and its length).
 - Operations (methods/functions) that operate on the data (e.g., `find()`, `lower()`, `endswith()`).
- Grouping data and methods together forms a class.

- Python's Built-in `str` Class

- Python provides a built-in class named `str` to represent strings.
- When you create a string, e.g., `s1 = 'Hello'`, you are creating an object from the `str` class.
- You can verify an object's class using `type()`, e.g., `type(s1)` returns `<class 'str'>`.

- Methods of the `str` Class

- The `str` class includes numerous methods like:
 - `find()`, `index()`, `count()`, `join()`, `endswith()`, `isalpha()`, `lower()`, `upper()`, and many more.
- Methods are accessed using the dot operator, e.g., `s1.find('e')`.
- IDEs like PyCharm and VSCode provide auto-completion lists of these methods.
- The Python built-in function `dir()` can list all attributes and methods of a class or an object, e.g., `dir(str)`.

- Future Lectures

- Upcoming lectures will focus on learning these string methods in detail, grouped by similarity.

Section 3: Example Code and Use Cases

Creating a string object

Checking the type of the object

```
s1 = 'Hello'

print(type(s1)) # Output: <class 'str'>
```

Explanation: This example shows that the variable `s1` holds an object of type `str`.

Accessing string methods using dot operator

```
s1 = 'Hello'
print(s1.find('e'))      # Examples of methods: find substring
print(s1.endswith('lo')) # Check if string ends with substring
print(s1.isalpha())      # Check if string is alphabetic
print(s1.lower())        # Convert string to lowercase
```

Explanation: These demonstrate some common string methods that operate on the string data.

Using dir() function to list all methods and attributes of the str class

```
print(dir(str))
```

Explanation: This will print a list of all methods and special attributes available for the string class, including both normal and underscore methods.

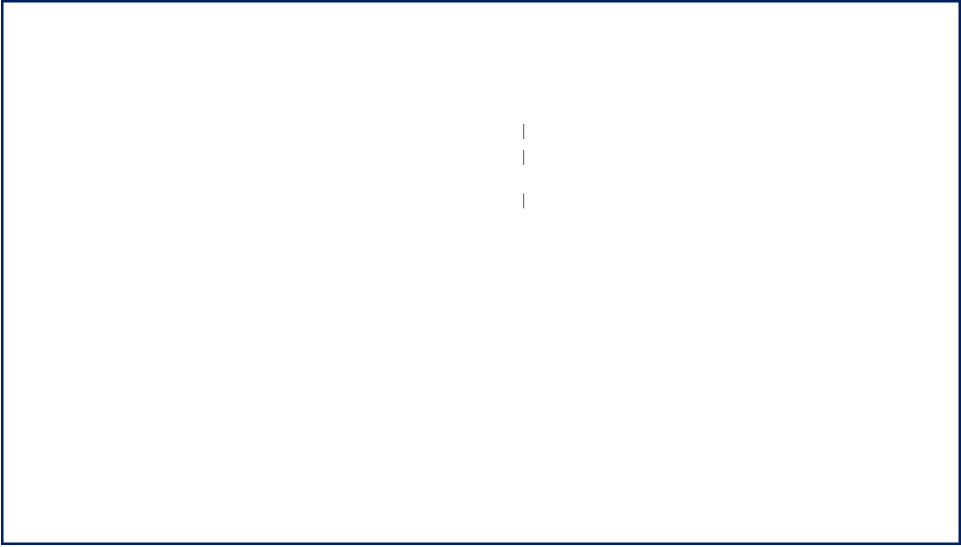
Section 4: Key Takeaways

- In Python, every string is an object belonging to the built-in `str` class.
- A class can be thought of as a blueprint that defines data and related operations (methods).
- Objects are individual instances created based on a class plan, each holding its own data.
- The `str` class includes numerous built-in methods that allow various string manipulations and checks.

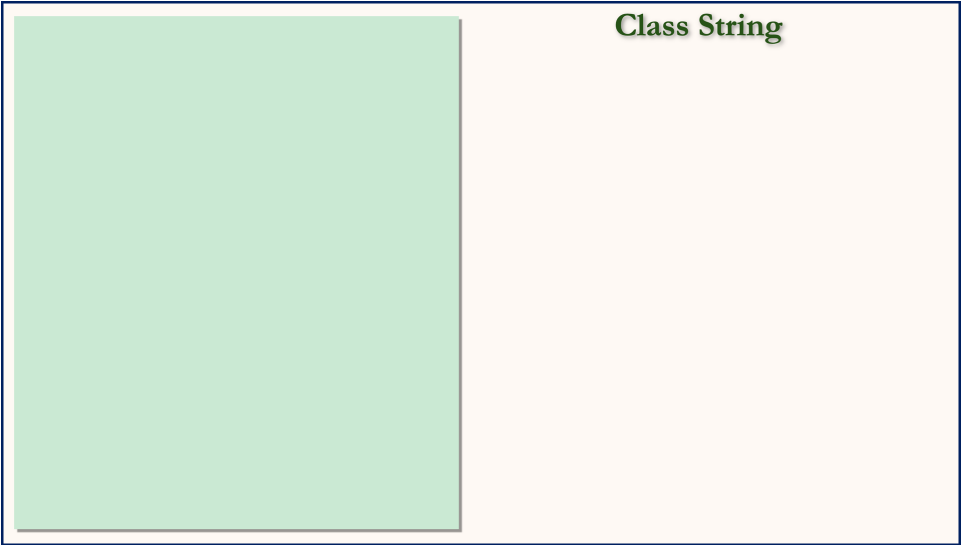
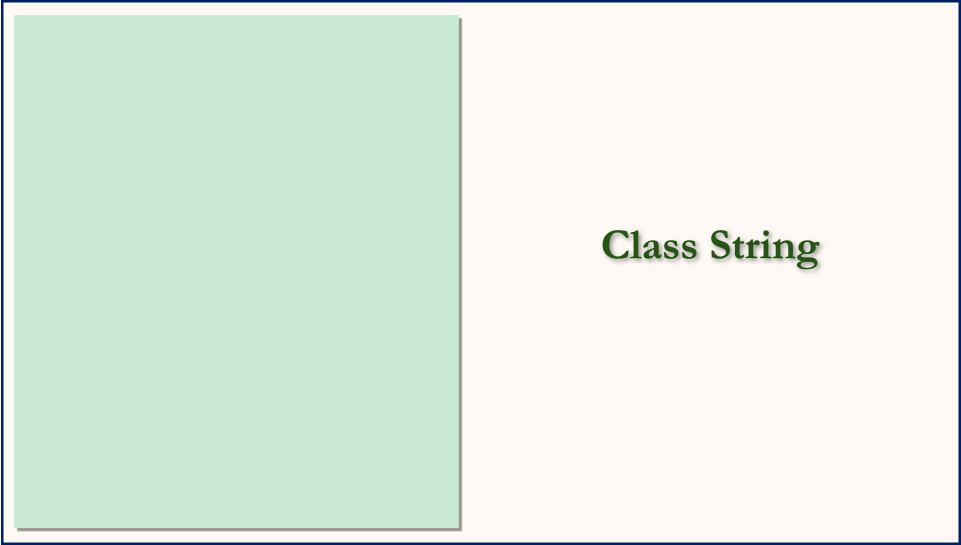
- Methods are accessed through the dot operator on string objects.
- IDEs and the `dir()` function can show all the available methods on string objects.
- Understanding classes and objects is crucial for using Python effectively, especially for complex data types like strings.
- Later lectures will delve into specific string methods to enhance string handling skills.

Lecture in Slides

Please Note: This document presents a curated selection of slides from the lecture; others have been excluded for conciseness.



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Class String

What is a String ?

Class String

What is a String ?

Class & Object

Class String

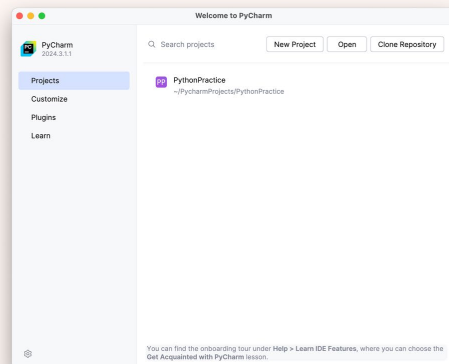
What is a String ?

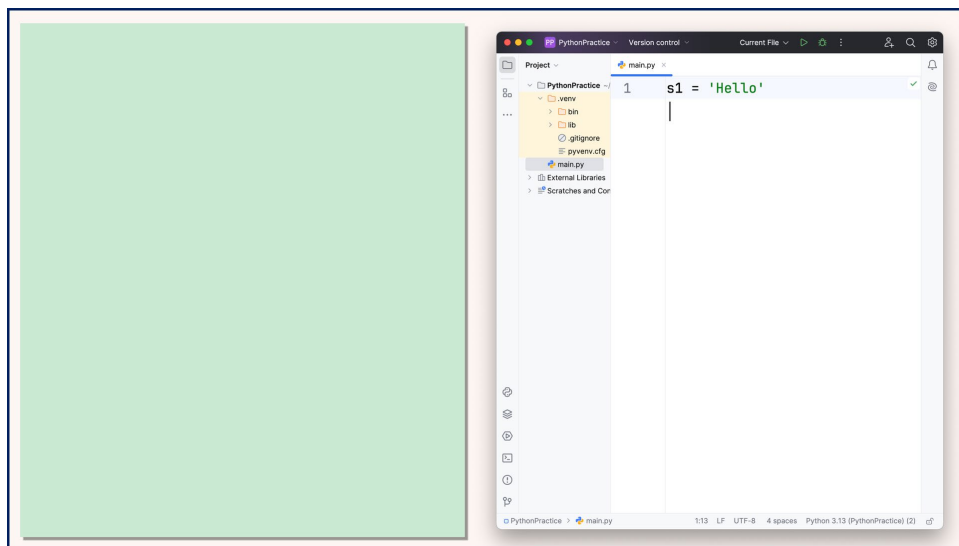
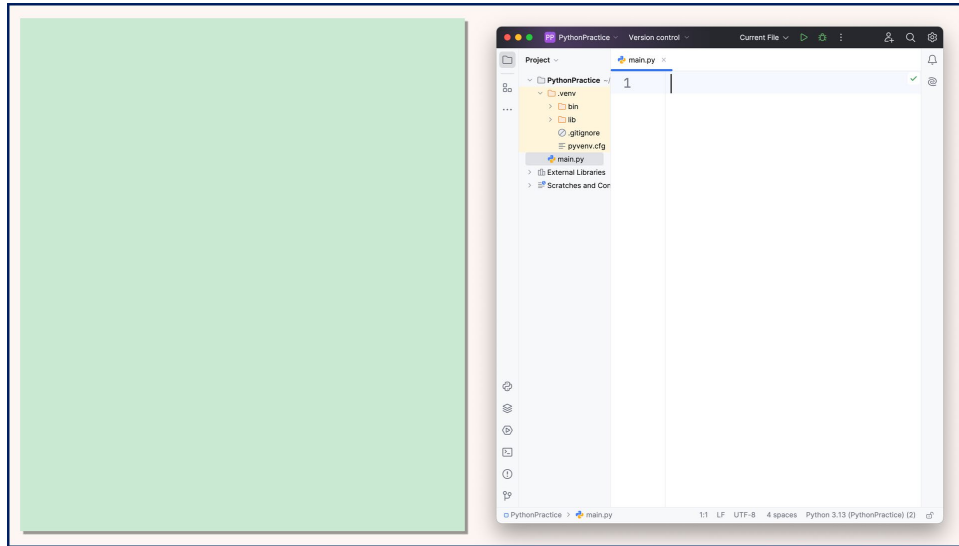
Class & Object

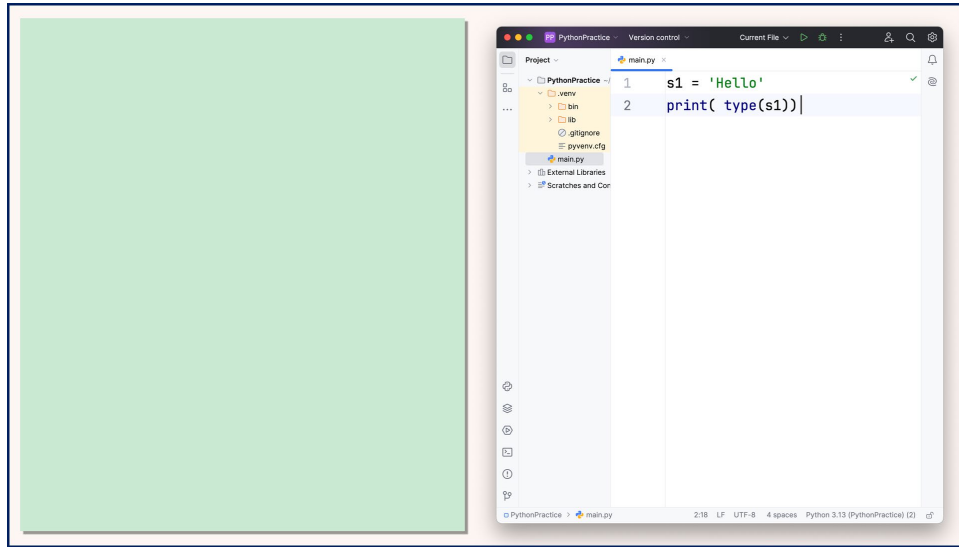
Methods of class str

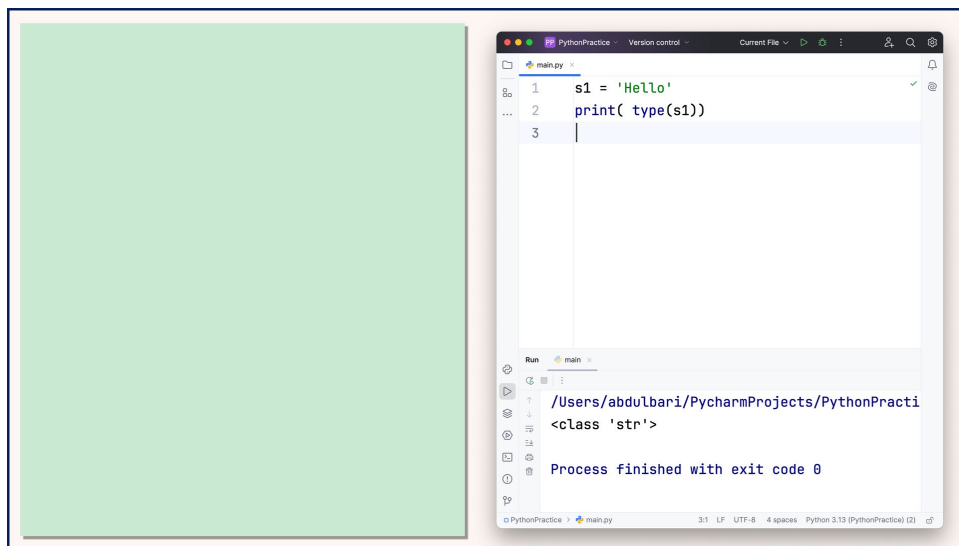
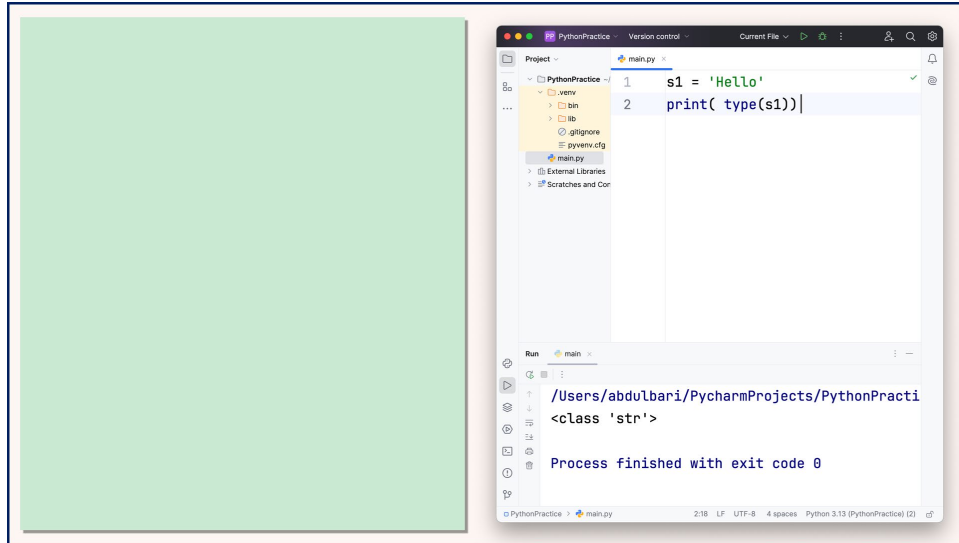
What is a String ?

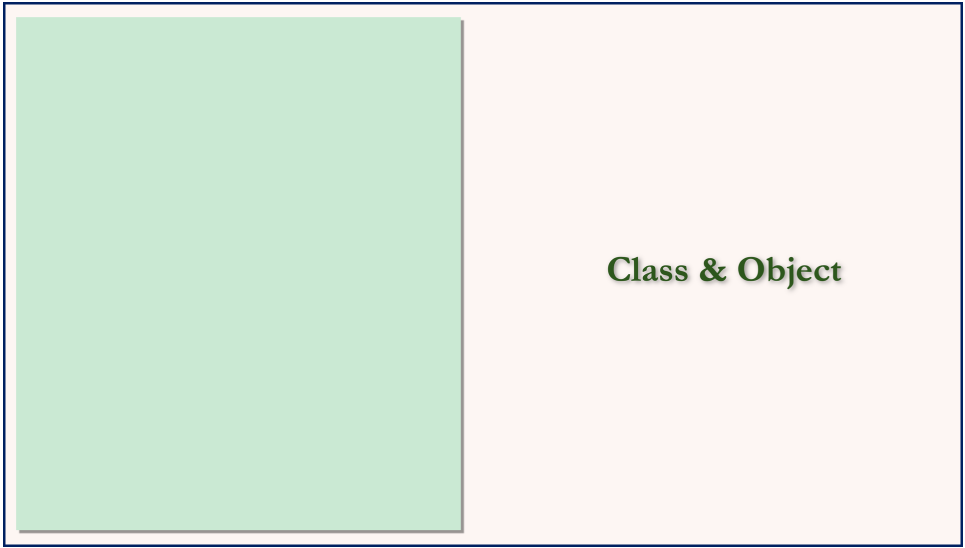
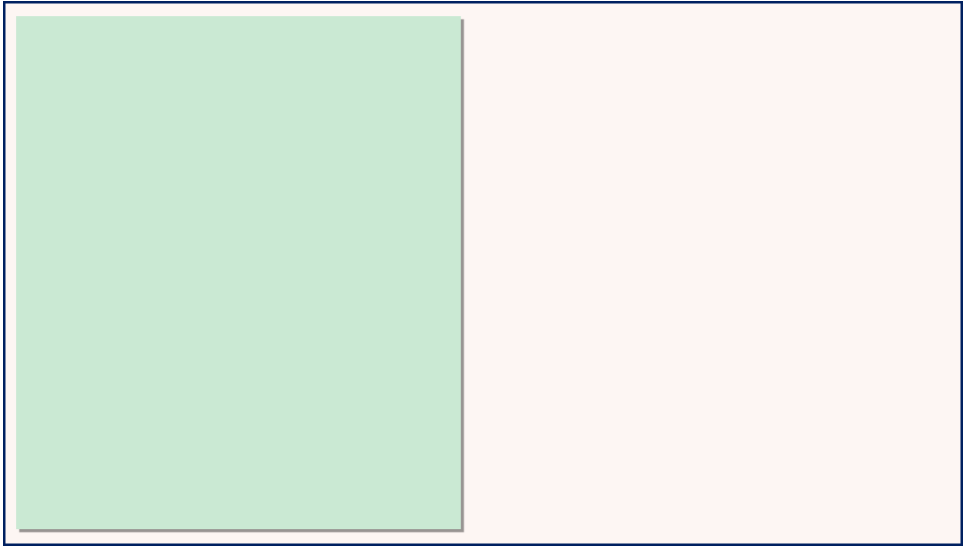
What is a String ?











Class & Object

Class & Object



Class & Object





Class & Object







Class & Object





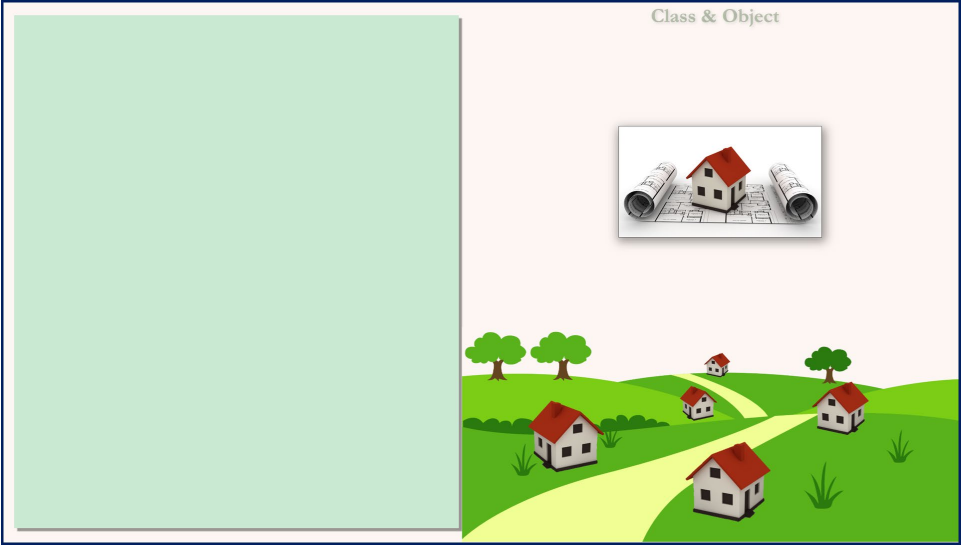
Class & Object





Class & Object





Class & Object

class str



Class & Object

class str

Data



Class & Object

class str

Data

Methods



Class & Object

class str

Data

string

Methods



Class & Object

class str

Data

string
length

Methods

Class & Object

class str

Data

string
length

Methods

find()

Class & Object

class str

Data

string

length

Methods

find()

index()

Class & Object

class str

Data

string

length

Methods

find()

index()

endswith()

Class & Object

class str

Data

string

length

Methods

find()

index()

endswith()

isalpha()



Class & Object

class str

Data

stringlength

Methods

find()
index()
endswith()
isalpha()
lower()





Class & Object

class str

Data

stringlength

Methods

find()
index()
endswith()
isalpha()
lower()
upper()





Class & Object

class str

Data	
string	
length	

Methods

find()	
index()	
endswith()	
startswith()	
lower()	
upper()	



Class & Object

class str

Data	
string	
length	

Methods

find()	
index()	
endswith()	
startswith()	
lower()	
upper()	

s1

Data

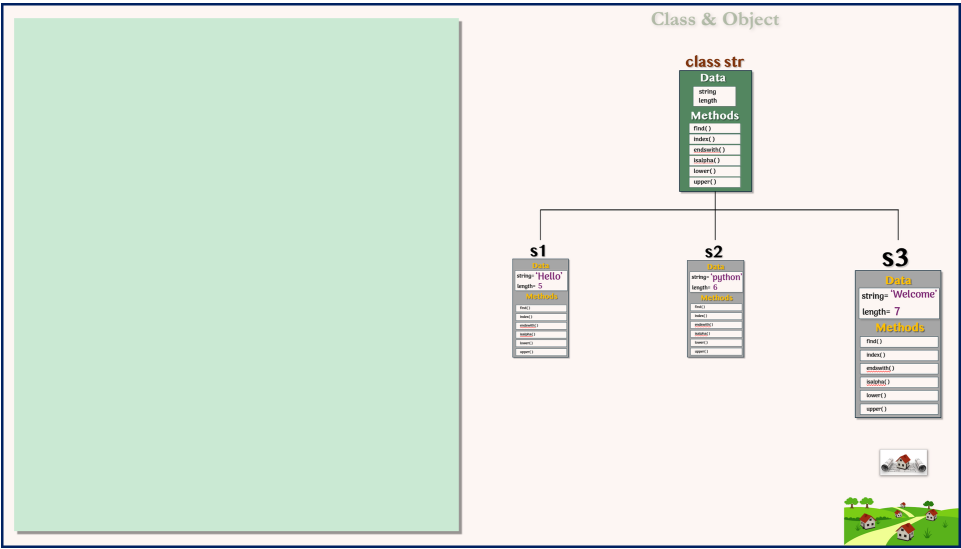
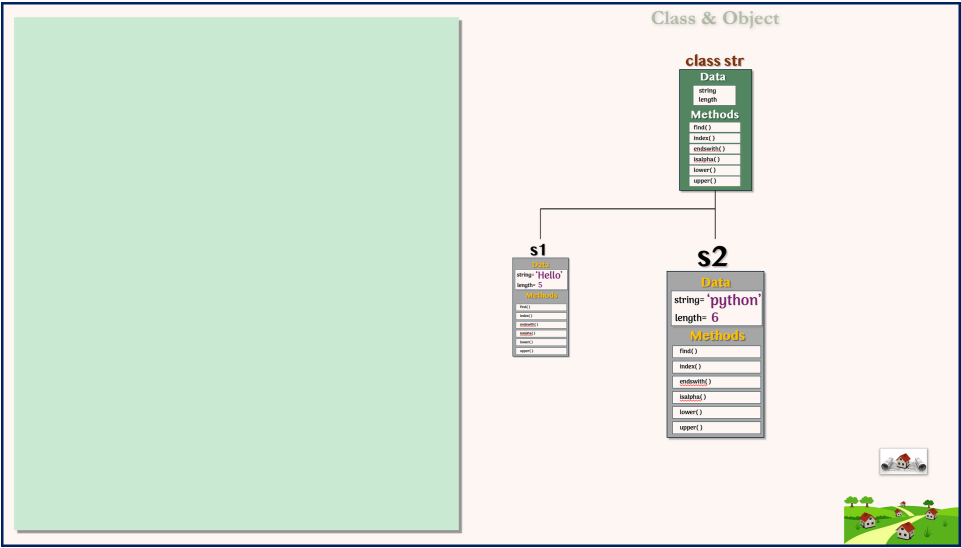
string= 'Hello'

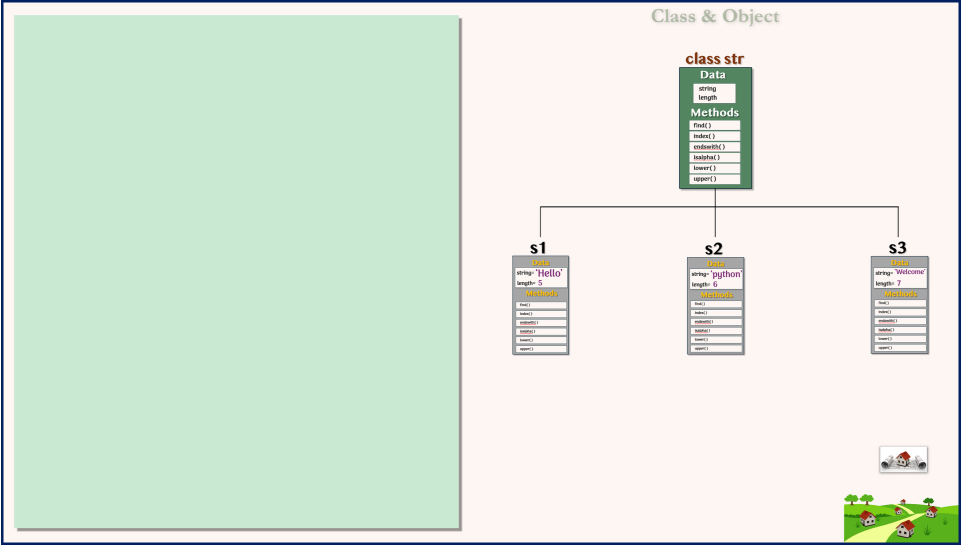
length= 5

Methods

find()	
index()	
endswith()	
startswith()	
lower()	
upper()	







Methods of class str

